

Rachel Chaiser

Department of Mathematics and Statistics
Vassar College
124 Raymond Ave
Poughkeepsie, NY 12603

Office: Rockefeller Hall 106B
rchaiser@vassar.edu

Education & Employment

Visiting Assistant Professor, Vassar College, Poughkeepsie, NY *Fall 2026–Present*

Visiting Assistant Professor, Kalamazoo College, Kalamazoo, MI *Fall 2024–Spring 2026*

University of Colorado Boulder, Boulder, CO

- Ph.D., Mathematics *May 2024*
- Dissertation Title: *A Low-Dimensional Counterexample to the HK-Conjecture*
- Advisor: Robin Deeley
- M.A., Mathematics *May 2022*

University of Puget Sound, Tacoma, WA

- B.S. with Honors in Mathematics, Minor in Education Studies *May 2018*
- Cum Laude, 3.79/4.0 Cumulative GPA, Outstanding Senior Award in Mathematics

Teaching

Vassar College *Fall 2026–Present*

Visiting Assistant Professor

Responsibilities: create syllabi; write assignments and assessments; develop active-learning worksheets; and create and maintain a learning management system (Moodle).

- Calculus II — *Fall 2026 (2 sections)*

Kalamazoo College *Fall 2024–Spring 2026*

Visiting Assistant Professor

Responsibilities: create syllabi; write assignments and assessments; develop active-learning worksheets; and create and maintain a learning management system (Moodle).

- Abstract Algebra I — *Spring 2026*
- Linear Algebra — *Fall 2025, Spring 2025*
- Calculus III — *Spring 2025*
- Calculus II — *Winter 2026, Spring 2026*
- Calculus I with Review Part 2 — *Winter 2025 (2 sections), Winter 2026*
- Calculus I with Review Part 1 — *Fall 2024 (2 sections), Fall 2025*

University of Colorado Boulder *Fall 2019–Summer 2024*

Instructor

Responsibilities: create and maintain a learning management system (Canvas); create syllabus, activities, assignments, and assessments.

- MATH 1212: Data and Models — *Summer 2024*
Course Description: Excel-based introductory statistics course.

Graduate Part Time Instructor

Responsibilities: create and maintain a learning management system (Canvas); create syllabus, activities, assignments, and assessments.

- Linear Algebra for Non-Mathematics Majors — *Spring 2023*

Graduate Teaching Assistant (Instructor of Record)

Responsibilities: lecture; write and grade quizzes and exams.

- Calculus III — *Spring 2024*
- Calculus II — *Fall 2021, Spring 2022*
- Calculus I — *Fall 2020, Spring 2021*

Graduate Teaching Assistant (TA)

Responsibilities: lead recitation; facilitate group work; grade homework, projects, and exams; tutor in the Mathematics Academic Resource Center.

- Calculus III — *Fall 2023*
- Calculus I — *Spring 2020*
- Precalculus — *Fall 2019*

Advanced Undergraduate Teaching

Experimental Math Lab

Fall 2022

Title: *Computing invariants for subshifts of finite type.*

Lectured on dynamical systems called shifts of finite type, their invariants and computational techniques.

Led bi-weekly problem sessions. Taught basic coding principles and computational techniques in SageMath.

REU/G

Summer 2021

Title: *Computing invariants for expansive dynamical systems.*

Mentored 4 undergraduate and 2 first-year graduate students in a 6-week research project on algebraic topology and dynamical systems. Lectured on invariants for shifts of finite type. Taught a mini-course on cellular homology and computational methods. Led 3x weekly problem sessions.

Summer Multicultural Access to Research Training Program (SMART)

Summer 2021

SMART is an outreach program that increases research opportunities for underrepresented groups. Partnering with the above REU/G, I mentored an undergraduate student for 10 weeks. I guided his independent research and the creation of his first research talk and poster presentation.

Publications

1. **R. Chaiser**, “A low-dimensional counterexample to the HK-conjecture”, submitted to Proceedings of the AMS, arXiv:2507.05425.
2. **R. Chaiser**, M. Coates-Welsh, R. J. Deeley, A. Farhner, J. Giornozi, R. Huq, L. Lorenzo, J. Oyola-Cortes, M. Reardon, and A. M. Stocker “Invariants for the Smale space associated to an expanding endomorphism of a flat manifold”, *Munster Journal of Mathematics* **16** (2023), 177-199.
3. **R. Chaiser**, K. Pelatt, C. Thatcher, “Using the particle model to find structure in Eilenberg–MacLane spaces”, *The PUMP Journal of Undergraduate Research* **4** (2021), 86-107.
4. T. Ball, **R. Chaiser**, D. Dustin, T. Edgar, P. Lagarde, “Some Combinatorics from Zeckendorf Representations”, *Involve* **12** (2019), No. 7, 1241-1260.

Conference Attendance

1. Invited. Applications of Operator Algebras and Noncommutative Geometry to Quantum Materials. Banff International Research Station, Alberta, Canada, Apr 11–16 2027.
2. Invited. Workshop on Noncommutative Geometry. University of Hawai'i at Manoa, Nov 27–29 2026.
3. Invited. From Dynamics to C^* -Algebras and Back Again: Exploring Topological Invariants at Brin MRC, University of Maryland, Oct 3–5 2026.
4. Noncommutative Geometry Festival. University of Colorado Boulder, Mar 1–2 2025.
5. MAA Rocky Mountain Section Meeting. Colorado College, Apr 19–21 2024.
6. Young Mathematicians in C^* -Algebras 2023. KU Leuven, Belgium, Aug 7–11, 2023.
7. Great Plains Operator Theory Symposium 2023. The Ohio State University, May 15–19, 2023.
8. Groupoidfest. University of Colorado Boulder, Nov 12–13 2022.
9. Young Mathematicians in C^* -Algebras 2022. University of Oslo, Norway, Aug 8-12, 2022.
10. Great Plains Operator Theory Symposium 2022. Washington University in St. Louis, May 23–28 2022.
11. Groupoidfest. University of Colorado Colorado Springs, Nov 13–14 2021.
12. Young Mathematicians in C^* -Algebras 2021. University of Münster, Germany, Aug 9-13, 2021.

Presentations

External Talks

1. “Introduction to K -Theory for C^* -Algebras, Part 1.” Mini-course lecture. Groundwork for Operator Algebras Lecture Series. July 14, 2025.
2. “A Low-Dimensional Counterexample to the HK-Conjecture.” Invited. Michigan State University Analysis Seminar. March 18, 2025.
3. “A Low-Dimensional Counterexample to the HK-Conjecture.” Invited. Noncommutative Geometry Festival. University of Colorado, Boulder. March 1, 2025.
4. “A Low-Dimensional Counterexample to the HK-Conjecture.” MAA Rocky Mountain Section Meeting. April 19, 2024.
5. “A Low-Dimensional Counterexample to the HK-Conjecture.” Invited. Arizona State University C^* -algebra Seminar. November 2, 2023.
6. “A Low-Dimensional Counterexample to the HK-Conjecture.” Invited. University of Hawaii Manoa Functional Analysis Seminar. October 23, 2023.
7. “Wieler Solenoids from Flat Manifolds Part 1: The Stable Groupoid.” Smale spaces, their groupoids and C^* -algebras, online conference. May 15, 2023.
8. “An Introduction to Kasparov’s KK -Theory.” Student Seminar, Groundwork for Operator Algebras Lecture Series. March 29, 2022.
9. “Morita Equivalence for C^* -algebras.” Student Seminar, Groundwork for Operator Algebras Lecture Series. January 28, 2021.
10. “Exploring Homology Through the Particle Model.” MAA MathFest 2018 Pi Mu Epsilon Student Paper Sessions. August 3, 2018.
11. “The Combinatorics of Zeckendorf Representations.” Nebraska Conference for Undergraduate Women in Mathematics. January 27, 2018.

12. With D. Dustin and P. Lagarde, “The Combinatorics of the Zeckendorf Representations of the Natural Numbers.” Joint Mathematics Meetings, MAA Undergraduate Poster Session. January 12, 2018.
13. “The Combinatorics of Zeckendorf Representations.” Department Seminar, Pacific Lutheran University. November 15, 2017.
14. “The Combinatorics of Zeckendorf Representations.” Northwest Undergraduate Mathematics Symposium. October 14, 2017.
15. With D. Dustin and P. Lagarde, “The Combinatorics of Zeckendorf Representations.” Colloquium for UWB and Seattle University REU researchers. University of Washington, Bothell. August 10, 2017.

Internal Talks

1. “An Introduction to K -Theory for Banach Algebras.” Topology Seminar, University of Colorado Boulder. November 7, 2023.
2. “Flat Manifolds and the HK-Conjecture.” Topology Seminar, University of Colorado Boulder. April 25, 2023.
3. “Flat Manifolds and the HK-Conjecture.” Functional Analysis Seminar, University of Colorado Boulder. March 16, 2023.
4. “ KK -Theory for C^* -Algebras: A Proof that $KK(\mathbb{C}, A)$ is Isomorphic to $K_0(A)$.” Functional Analysis Seminar, University of Colorado Boulder. April 14, 2022.
5. “ KK -Theory for C^* -Algebras: Defining $KK(A, B)$.” Functional Analysis Seminar, University of Colorado Boulder. March 3, 2022.
6. “An Introduction to C^* -algebras.” Graduate Student Seminar, University of Colorado Boulder. November 19, 2020.
7. “Morita Equivalence.” Functional Analysis Seminar, University of Colorado Boulder. November 5, 2020.
8. “Hilbert C^* -modules - Projections and Unitaries.” Functional Analysis Seminar, University of Colorado Boulder. October 1, 2020.
9. “Exploring Homology Through the Particle Model.” University of Puget Sound Math/CS Day conference. April 28, 2018.
10. “Eight Weeks of Zeckendorf Representations.” Department Seminar, University of Puget Sound. February 19, 2018.

Service

University of Colorado Boulder

Graduate Student Representative to the Graduate Committee

Fall 2021–Spring 2023

Acted as a liaison between graduate students and faculty leadership. Negotiated graduate student interests and collaborated with students and faculty to provide solutions to student grievances.

Mathematics Department Diversity Committee

Fall 2019–Spring 2023

Partnered with the SMART program in Summer 2021 (see Teaching section). Organized the following two mentor programs.

Mathematics Department Prelim Mentoring Program

Organizer

Fall 2022–Summer 2023

Wrote and received a \$7k grant through the Dean’s Innovation Fund to fund the program. Hired and trained mentors for each preliminary exam. Designed and organized the program.

Mathematics Department Peer Mentoring Program

Organizer

Fall 2021–Spring 2024

Selected and trained graduate student mentors, paired mentors with first-year mentees, advertised the mentor program, solicited and incorporated mentor and mentee feedback.

Mentor

Fall 2020–Spring 2023

Partnered with another graduate student to mentor 2 first-year graduate students. Met monthly to discuss how to succeed in graduate school, give advice and support.

Department Tea Organizer

Fall 2021–Spring 2022

Restarted our monthly department tea post-COVID. Bought snacks and refreshments, organized games like crosswords and chess, navigated reimbursement.

Groundwork for Operator Algebras Lecture Series (GOALS)

Teaching Assistant

Summer 2022

Wrote problem sets, led problem sessions, and organized community building events and excursions for this graduate student summer school focused on C^* -algebras and von Neumann algebras.

Honors and Awards

Burton Jones Teaching Excellence Award

AY 2023–2024

Awarded to a veteran graduate teaching assistant or graduate part-time instructor in the Department of Mathematics at the University of Colorado at Boulder in recognition of outstanding accomplishments in teaching.

CU Boulder Graduate School Summer Fellowship

Summer 2023

Dean's Innovation Fund

Spring 2023

Awarded \$7,000 to launch a mentor program for graduate students studying for their preliminary exams.

Edward G. Goman Mathematics / Computer Science Outstanding Senior Award

Spring 2018

Awarded to the top graduating senior in the Mathematics and Computer Science Department at the University of Puget Sound.

Outreach

Summer Camp Teaching Assistant

Summer 2023

University of Colorado Science Discovery

Supported daily instruction of coding fundamentals and linear algebra using the game development engine *Godot* for 15 high school students; culminating in the development of their own original video game.

Mathematics Tutor

Nov 2018–Jun 2019

Lincoln High School, Tacoma, WA

Launched an afterschool tutoring program at a local under-resourced high school for students earning at most a C in math, or at risk of not graduating due to falling below the district's mathematics standards.

Camp Counselor

Jul 2018

MathPath

Supervised a group of 9 highly motivated middle school girls. Helped foster a culture of respect, collaboration, and academic engagement by showing respect for all others, and by encouraging discussion, community, and persistence in scholarship. Assisted in the daily teaching advanced topics, while observing veteran teachers.

Volunteer Mathematics Tutor*Sep 2015–May 2017**Access Programs at the University of Puget Sound*

Provided homework help and supplemental instruction for 7th-12th graders in this program targeted at increasing access to college for traditionally underrepresented groups; 2–4 hours per week. Mentored students and gave advice for selecting and applying for colleges.

Mathematics Tutor at a Women's Prison*Fall 2016**Washington Corrections Center for Women*

Provided weekly supplemental instruction for incarcerated women as the experiential learning component of the course REL 307: PRISON, GENDER, AND EDUCATION. This work supported the establishment of the Freedom Education Project Puget Sound, which supports re-integration by providing a path to earn an AA or BA.

Summer Academic Challenge Teaching Assistant*June 2016–July 2016**Access Programs at the University of Puget Sound*

Supported daily instruction of a STEM curriculum built around space travel and the solar system for rising 9th and 10th graders. Planned and taught a lesson where students experimented with model rocket designs and used trigonometry to gauge which design reached the highest altitude. Supervised students in collaborative groups of 4–5 in performing library research, writing a research paper, and creating and giving presentations.